Fourier series representation of discrete-time periodic signals:

The first question we need to ask is: what is a discrete-time periodic signal.

⊡ A discrete signal $x[n]$ is periodic with period N if $x[n] = x[n+N]$, where N is a positive integer.

→ The fundamental period is the smallest positive integer N for which equation ① holds.

→ The fundamental frequency is $\Omega = 2\pi/N$.

e.g. $x[n] = e^{j(2\pi/N)n}$ is a periodic signal.

Note that $x[n+N] = e^{j(2\pi/N)(N+n)}$

$$= e^{j2\pi} \cdot e^{j(2\pi/N)n} = e^{j(2\pi/N)n} = x[n]$$

$\therefore e^{j(2\pi/N)n}$ is a periodic signal of period N.

Now like the continuous-time case we look into what does "harmonics" mean for the discrete case.

For a continuous signal like $x(t) = e^{j\frac{2\pi}{T}t}$, the harmonics are $x_k(t) = e^{j\,2\pi/T\,kt}$, $k \in \{2, 3, \dots\}$.

In a similar way, if we write the harmonics of the discrete-time signal $e^{j\,2\pi/N\,n} = x[n]$, we get

$$x_k[n] = e^{j(2\pi/N)kn}, \quad k \in \{2, 3 \dots\}.$$

(Here $k = 1$ is the fundamental and the rest its harmonics)

Unlike continuous-time case note that for the discrete-time case we only have $(N-1)$ distinct harmonics of $e^{j\,2\pi/N\,n}$. These $(N-1)$ distinct harmonics are:

$$e^{j\,2\pi/N\,n} \longrightarrow k = 1 \text{ is the fundamental.}$$

$$\left.\begin{array}{ll} k = 2: & e^{j\,2\pi/N\,2n} \\ k = 3: & e^{j\,2\pi/N\,3n} \\ & \vdots \\ k = N-1: & e^{j\,2\pi/N\,(N-1)n} \\ k = N: & e^{j\,2\pi/N\,\cdot\,Nn} \longrightarrow k = 0 \text{ case} \end{array}\right\} (N-1) \text{ harmonics.}$$

Note for $k = N+1$, we get

$$e^{j(2\pi/N)(N+1)n} = e^{j2\pi n} \cdot e^{j2\pi/N\, n}$$

$$= \underbrace{\left[\cos(2\pi n) + j\sin(2\pi n)\right]}_{= 1 \text{ for all } n \in \mathbb{N}} e^{j2\pi/N\, n}$$

$$= e^{j2\pi/N\, n} \quad \longleftarrow \quad k = 1, \text{ fundamental signal.}$$

Similarly, $e^{j\left(\frac{2\pi}{N}\right)(N+2)n} = e^{j(2\pi/N)\,2n} \leftarrow k = 2$ case

$\therefore$ The only distinct harmonics of $e^{j\frac{2\pi}{N}n}$ are

$$e^{j\frac{2\pi}{N}2n}, \quad e^{j\frac{2\pi}{N}\cdot 3n} \quad \ldots\ldots \quad e^{j\frac{2\pi}{N}\cdot Nn}$$

$$\boxed{\text{A continuous signal } e^{j\frac{2\pi}{T}t} \text{ has infinite harmonics: } e^{j2\pi/T\, 2t}, \quad e^{j2\pi/T\, 3t}, \ldots\ldots}$$

Thus, analogous to the continuous-time case we wish to represent the discrete-time periodic signal as sum of cosines. Such linear combination has the form:

$$x[n] = \sum_k a_k e^{j(2\pi/N)kn} \quad \left[\text{Here } x[n] \text{ has period } N\right]$$

$$\therefore \quad x[n] = \sum_{k} a_k e^{j\left(\frac{2\pi}{N}\right)kn}$$

$$= \sum_{k=1}^{N} a_k e^{j(2\pi/N)kn} \longrightarrow (1)$$

This can be written because $e^{j(2\pi/N)(N+k)n} = e^{j\left(\frac{2\pi}{N}\right)kn}$

There is another notation that is generally used.

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{j(2\pi/N)kn} \longrightarrow (2)$$

This is the modulo symbol.

This means $k \in \{0, 1, \ldots N-1\}$

This is same as the $k = N$ case. so the sum in eqn (1) & (2) are the same.

What is 'modulo' operation?

$\langle 2 \rangle$ means mod 2 operation.

i.e. if $k = 3$ then $[3/2] = 1$ is the remainder.

$\therefore$ $3 \bmod 2 = 1$. (We are taking the remainder instead of the quotient)

$\therefore$ When we write $k = \langle N \rangle$ this means

$k = N$ becomes 0 (N/N remainder is 0)

$k = N+1$ becomes 1 ($N+1/N$ remainder is 1) .....

$\therefore$ $k \in \{0, 1, 2, \ldots N-1\}$

$\therefore$ The Fourier-series is

$$x[n] = \sum_{k=\langle N \rangle} a_k \, e^{j\left(\frac{2\pi}{N}\right)kn} \longrightarrow (2)$$

Fourier-series co-efficients / Spectral co-efficients.

The general questions now are:

(a) Is it always possible to write a signal using equation (2)?

(b) How do we find the spectral coefficients?

From equation ② we can write —

N-number of linear equations with N unknowns $a_0, a_1, \ldots, a_{N-1}$.

$$\begin{cases} x[0] = \sum_{k=0}^{N-1} a_k = a_0 + a_1 + \cdots + a_{N-1} \\ x[1] = \sum_{k=0}^{N-1} a_k e^{j(2\pi/N)k} \\ \quad = a_0 + a_1 e^{j(2\pi/N)} + \cdots + a_{N-1} e^{j\left(\frac{2\pi}{N}\right)(N-1)} \\ x[2] = \sum_{k=0}^{N-1} a_k e^{j(2\pi/N)2k} \\ \quad = a_0 + a_1 e^{j(2\pi/N)2} + \cdots + a_{N-1} e^{j(2\pi/N)2(N-1)} \\ \quad \vdots \\ x[N-1] = a_0 + a_1 e^{j\left(\frac{2\pi}{N}\right)(N-1)} + \cdots + a_{N-1} e^{j\left(\frac{2\pi}{N}\right)(N-1)^2} \end{cases}$$

$x[N] = x[0]$ ⟵ This is because $x[N]$ has period N.

So, if we can solve the above N equations with N unknowns we will get $a_0, a_1, \ldots, a_{N-1}$. Other points $x[N], x[N+1], \ldots$ and $x[-1], x[-2], \ldots$ are not required

because $x[-1] = x[N-1]$, $x[N] = x[0]$

$x[-2] = x[N-2]$, $x[N+1] = x[2]$ ....

$\vdots$

We write all the equations in matrix form:

$$\begin{bmatrix} x[0] \\ x[1] \\ \vdots \\ x[N-1] \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & e^{j(2\pi/N)} & \cdots & e^{j\left(\frac{2\pi}{N}\right)(N-1)} \\ \vdots & & & \\ 1 & e^{j\left(\frac{2\pi}{N}\right)(N-1)} & \cdots & e^{j\left(\frac{2\pi}{N}\right)(N-1)^2} \end{bmatrix}}_{M \text{ (say)}} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_{N-1} \end{bmatrix}$$

It can be shown that M is invertible using the idea that M is a Vandermonde matrix with N distinct non-zero eigenvalues. The proof is not in the scope of this course.

Thus,

$$\begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_{N-1} \end{bmatrix} = M^{-1} \begin{bmatrix} x[0] \\ x[1] \\ \vdots \\ x[N-1] \end{bmatrix} \longrightarrow (3)$$

spectral-coefficients.

e.g. $x[n] = \sin\left(\frac{2\pi}{N}\right)n$, where $N = 3$ $\longrightarrow$ (4)

Here $x[n]$ is a periodic signal with period

$\therefore$ Fourier-series will be:

$$x[n] = \sum_{k=\langle 3 \rangle} a_k e^{j\left(\frac{2\pi}{3}\right)kn}$$

$$= \sum_{k=0}^{2} a_k e^{j\left(\frac{2\pi}{3}\right)kn}$$

From (4), we have $x[0] = 0$

$$x[1] = \sin\left(\frac{2\pi}{3}\right) = \sqrt{3}/2$$

$$x[2] = \sin\left(\frac{4\pi}{3}\right) = -\sqrt{3}/2$$

$$\therefore \begin{bmatrix} x[0] \\ x[1] \\ x[2] \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & e^{j(2\pi/3)} & e^{j(2\pi/3)2} \\ 1 & e^{j(2\pi/3)2} & e^{j(2\pi/3)4} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 0 \\ \sqrt{3}/2 \\ -\sqrt{3}/2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & e^{j(2\pi/3)} & e^{j(4\pi/3)} \\ 1 & e^{j(4\pi/3)} & e^{j(8\pi/3)} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1/2 + \sqrt{3}/2\, j & -1/2 - \sqrt{3}/2\, j \\ 1 & -1/2 - \sqrt{3}/2\, j & -1/2 + \sqrt{3}/2\, j \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ \sqrt{3}/2 \\ -\sqrt{3}/2 \end{bmatrix}$$

$$\therefore \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 0 \\ -1/2\, j \\ 1/2\, j \end{bmatrix}$$

$$\therefore \quad x[n] = -1/2\, j\, e^{j(2\pi/3)n} + 1/2\, j\, e^{j(2\pi/3)2n}$$

<u>Alternate way of finding spectral co-efficients</u> :

Recall that for continuous-time periodic signal there is a closed form expresion to compute the spectral co-efficients. This expresion is

$$a_k = \frac{1}{T} \int_T x(t)\, e^{-j\frac{2\pi}{T}kt}\, dt$$

For discrete-time we are using linear-equations to find the spectral-coefficients. Is there a closed form expression like that in case of continuous-time?

YES WE CAN! The formula is —

$$\boxed{a_r = \sum_{n=\langle N \rangle} x[n] e^{-jr\frac{2\pi}{N}n}}$$

<u>Proof:</u>

$$\sum_{n=\langle N \rangle} x[n] e^{-jr\frac{2\pi}{N}n}$$

$$= \sum_{n=\langle N \rangle} \left(\sum_{k=\langle N \rangle} a_k e^{jk\frac{2\pi}{N}n} \right) e^{-jr\frac{2\pi}{N}n}$$

$$= \sum_{n=\langle N \rangle} \sum_{k=\langle N \rangle} a_k e^{j(k-r)\frac{2\pi}{N}n}$$

$$= \sum_{k=\langle N \rangle} a_k \underbrace{\sum_{n=\langle N \rangle} e^{j(k-r)\frac{2\pi}{N}n}}_{\text{We analyze this expression next}} \longrightarrow \text{A}$$

Let $k - r =: \Delta$. Then we analyze the expression —

$$\sum_{n=\langle N\rangle} e^{j\Delta \frac{2\pi}{N} n}$$

Let $\Delta = pN$ then

$$\sum_{n=\langle N\rangle} e^{j(pN)\cdot\left(\frac{2\pi}{N}\right)\cdot n} = \sum_{n=\langle N\rangle} e^{j2\pi\cdot pn} = \sum_{n=\langle N\rangle} 1 = N.$$

For $\Delta \neq pN$, we use the finite sum formula for G.P.

$$\sum_{n=\langle N\rangle} e^{j\Delta \frac{2\pi}{N} n} = \frac{e^{j\Delta(2\pi/N)\cdot 0}\left(1 - e^{j\Delta(2\pi/N)N}\right)}{1 - e^{j\Delta(2\pi/N)}}$$

$$\left[\begin{array}{ll} a = e^{j\Delta(2\pi/N)\cdot 0}, & \text{No. of terms} = N \\ r = e^{j\Delta(2\pi/N)}, & \text{G.P. sum formula} = \frac{a(1-r^n)}{1-r} \end{array}\right]$$

$$= \frac{1 - \underbrace{e^{j\Delta 2\pi}}_{\to 1}}{1 - e^{j\Delta(2\pi/N)}} = 0.$$

$$\therefore \sum_{n=\langle N\rangle} e^{j\Delta\left(\frac{2\pi}{N}\right)n} = \begin{cases} N, & \text{for } \Delta = 0, N, 2N \ldots \\ 0, & \text{otherwise} \end{cases}$$

$$\therefore \text{(A)} \Rightarrow \sum_{n=\langle N\rangle} x[n]\, e^{-jr\left(\frac{2\pi}{N}\right)n} = a_r N \quad \left(\because \Delta = 0 \Rightarrow k = r\right)$$

$$\therefore \boxed{a_r = \frac{1}{N} \sum_{n=\langle N\rangle} x[n]\, e^{-jr\left(\frac{2\pi}{N}\right)n}} \rightarrow (5)$$

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Let us use equation (5) to solve the previous example.

We have $x[n] = \sin\left(\frac{2\pi}{3}\right)n \Rightarrow$ $x[0] = 0$, $x[1] = \sqrt{3}/2$, $x[2] = -\sqrt{3}/2$

$\therefore$ From equation (5):

$$a_0 = \frac{1}{3} \sum_{n=0}^{2} x[n]\, e^{0}$$

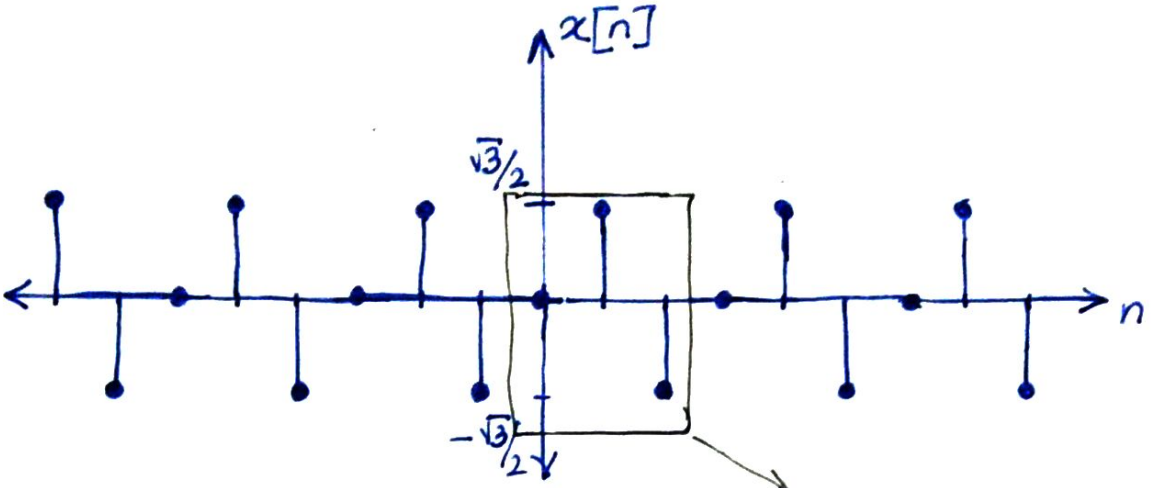
$$= \frac{1}{3}\left(x[0] + x[1] + x[2]\right) = 0$$

$$a_1 = \frac{1}{3}\left(x[0] + x[1]\, e^{-j\left(\frac{2\pi}{3}\right)} + x[2]\, e^{-j(2\pi/3)2}\right)$$

$$= -1/2\, j$$

$$a_2 = \frac{1}{3}\left(x[0] + x[1]\, e^{-j(4\pi/3)} + x[2]\, e^{-j(2\pi/3)4}\right) = \frac{1}{2} j$$

Note that pictorially the signal $x[n] = \sin\left(\frac{2\pi}{N}\right)n$, $N=3$ looks like this:



We have used information of only one period to compute the spectral co-efficient. What about the rest?

Rest won't be required. The spectral co-efficients will repeat itself

Note that

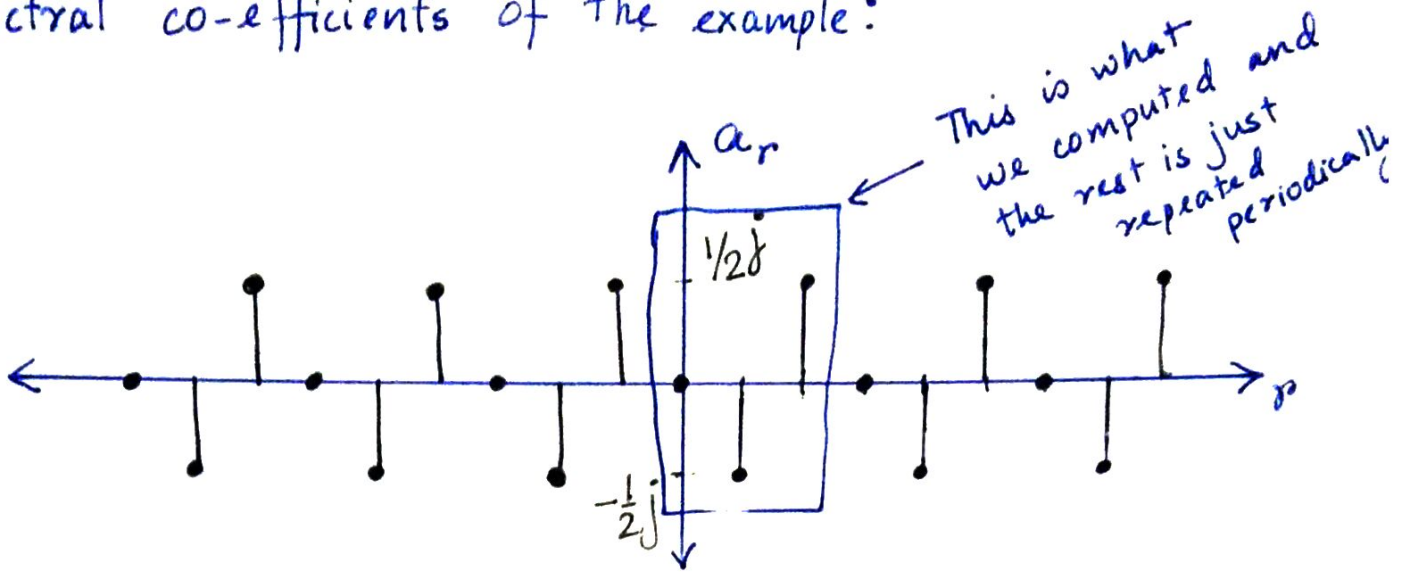
$$a_{r+N} = \frac{1}{N}\sum_{n=\langle N\rangle} x[n]\, e^{-j(2\pi/N)n(r+N)}$$

$$= \frac{1}{N}\sum_{n=\langle N\rangle} x[n]\, e^{-j\left(\frac{2\pi}{N}\right)n.r} \cdot e^{-j2\pi n}$$

$$= \frac{1}{N}\sum_{n=\langle N\rangle} x[n]\, e^{-j(2\pi/N)nr} = a_r$$

$\therefore$ The spectral co-efficients are also periodic with period N. We plot the

spectral co-efficients of the example:



→ <u>Convergence of Fourier-series for the discrete-case</u>

Recall that in the continuous case we had to deal with the convergence of Fourier-series. We found that for a square wave the finite sum

$$x(t) = \sum_{k=-N}^{N} a_k e^{j\frac{2\pi}{T}kt} \longrightarrow (6)$$

converges to the signal $x(t)$ as $N \to \infty$. However at the discontinuity we observed the Gibbs phenomenon, whereby, as the number of terms in eq^n (6) increased, the ripples in the partial sum (6) became compressed toward the discontinuity, with the peaks of the ripples remaining constant

independenty of the number of terms in the partial sum. The question therefore is — what about Gibb's phenomenon in discrete time case?

In case of discrete-time signals we donot have convergence issues. — this is because any discrete signal can be written as a <u>finite-sum</u> of exponentials. The Fourier-series is no longer infinite like that in the continuous case. So, there are no convergence issues and no Gibb's phenomenon.